

DBMS — Lecture 3

Network Model & Relational Data Model

Learning Objectives

Understand the network data model.

Understand the relational model and its basic terminology.

Compare network and relational models.

Network Model

The **network model** represents data as records connected through set types or links.

It supports complex many-to-many relationships and can be viewed as a graph-like structure.

Its navigation is often pointer/link based.

Relational Model

The **relational model** represents data using tables (relations).

A table consists of **rows (tuples)** and **columns (attributes)**.

A primary key uniquely identifies rows, while foreign keys represent relationships between tables.

Example

STUDENT(Student_ID, Name, Branch)

COURSE(Course_ID, Course_Name)

ENROLLMENT(Student_ID, Course_ID, Semester)

Comparison

Feature	Network Model	Relational Model
Representation	Records and links	Tables
Relationship	Links/set types	Keys and foreign keys
Access	Often navigational	Declarative queries such as SQL
Design simplicity	More complex	Generally simpler

Key Takeaway

The network model uses linked records, while the relational model organizes data into tables.